

Investment Decision Rules

NOTATION

r	discount rate
NPV	net present value
IRR	internal rate of return
PV	present value
$NPER$	annuity spreadsheet notation for the number of periods or dates of the last cash flow
$RATE$	annuity spreadsheet notation for interest rate
PMT	annuity spreadsheet notation for cash flow

IN 2017, AMAZON PURCHASED WHOLE FOODS FOR \$13.7 BILLION, by far the largest single investment decision the firm has ever made. Besides the up-front cost of the purchase, Amazon planned to spend significant resources integrating Whole Foods's bricks and mortar business into its online retail business. Presumably, Amazon executives believed that the acquisition would generate large synergies that would translate into greater future revenues. How did they know that the value of their future profits would exceed the significant investment required? More generally, how do firm managers make decisions they believe will maximize the value of their firms?

As we will see in this chapter, the NPV investment rule is the decision rule that managers should use to maximize firm value. Nevertheless, some firms use other techniques to evaluate investments and decide which projects to pursue. In this chapter, we explain several commonly used techniques—namely, the payback rule and the internal rate of return rule. We then compare decisions based on these rules to decisions based on the NPV rule and illustrate the circumstances in which the alternative rules are likely to lead to bad investment decisions. After establishing these rules in the context of a single, stand-alone project, we broaden our perspective to include deciding among alternative investment opportunities. We conclude with a look at project selection when the firm faces capital or other resource constraints.

7.1 NPV and Stand-Alone Projects

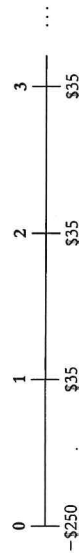
We begin our discussion of investment decision rules by considering a take-it-or-leave-it decision involving a single, stand-alone project. By undertaking this project, the firm does not constrain its ability to take other projects. To analyze such a decision, recall the NPV rule:

NPV Investment Rule: *When making an investment decision, take the alternative with the highest NPV. Choosing this alternative is equivalent to receiving its NPV in cash today.*

In the case of a stand-alone project, we must choose between accepting or rejecting the project. The NPV rule then says we should compare the project's NPV to zero (the NPV of doing nothing) and accept the project if its NPV is positive.

Applying the NPV Rule

Researchers at Fredrick's Feed and Farm have made a breakthrough. They believe that they can produce a new, environmentally friendly fertilizer at a substantial cost savings over the company's existing line of fertilizer. The fertilizer will require a new plant that can be built immediately at a cost of \$250 million. Financial managers estimate that the benefits of the new fertilizer will be \$35 million per year, starting at the end of the first year and lasting forever, as shown by the following timeline:



As we explained in Chapter 4, the NPV of this perpetual cash flow stream, given a discount rate r , is

$$NPV = -250 + \frac{35}{r} \quad (7.1)$$

The financial managers responsible for this project estimate a cost of capital of 10% per year. Using this cost of capital in Eq. 7.1, the NPV is \$100 million, which is positive. The NPV investment rule indicates that by making the investment, the value of the firm will increase by \$100 million today, so Fredrick's should undertake this project.

The NPV Profile and IRR

The NPV of the project depends on the appropriate cost of capital. Often, there may be some uncertainty regarding the project's cost of capital. In that case, it is helpful to compute an NPV profile: a graph of the project's NPV over a range of discount rates. Figure 7.1 plots the NPV of the fertilizer project as a function of the discount rate, r .

Notice that the NPV is positive only for discount rates that are less than 14%. When $r = 14\%$, the NPV is zero. Recall from Chapter 4 that the internal rate of return (IRR) of an investment is the discount rate that sets the NPV of the project's cash flows equal to zero. Thus, the fertilizer project has an IRR of 14%.

The IRR of a project provides useful information regarding the sensitivity of the project's NPV to errors in the estimate of its cost of capital. For the fertilizer project, if the cost of capital estimate is more than the 14% IRR, the NPV will be negative, as shown in

FIGURE 7.1

NPV of Fredrick's Fertilizer Project

The graph shows the NPV as a function of the discount rate. The NPV is positive only for discount rates that are less than 14%, the internal rate of return (IRR). Given the cost of capital of 10%, the project has a positive NPV of \$100 million.

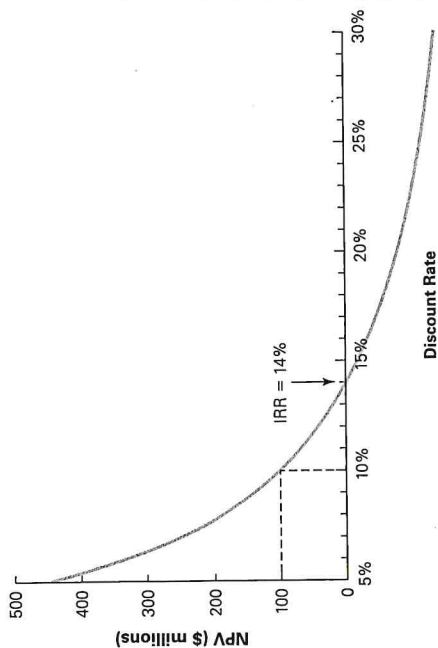


Figure 7.1. Therefore, the decision to accept the project is correct as long as our estimate of 10% is within 4% of the true cost of capital. In general, the difference between the cost of capital and the IRR is the maximum estimation error in the cost of capital that can exist without altering the original decision.

Alternative Rules Versus the NPV Rule

Although the NPV rule is the most accurate and reliable decision rule, in practice a wide variety of tools are applied, often in tandem with the NPV rule. In a 2001 study, 75% of the firms John Graham and Campbell Harvey¹ surveyed used the NPV rule for making investment decisions. This result is substantially different from that found in a similar study in 1977 by L. J. Gitman and J. R. Forrester,² who found that only 10% of firms used the NPV rule. MBA students in recent years must have been listening to their finance professors! Even so, Graham and Harvey's study indicates that one-fourth of U.S. corporations do not use the NPV rule. Exactly why other capital budgeting techniques are used in practice is not always clear. However, because you may encounter these techniques in the business world, you should know what they are, how they are used, and how they compare to NPV.

As we evaluate alternative rules for project selection in subsequent sections, keep in mind that sometimes other investment rules may give the same answer as the NPV rule, but at other times they may disagree. When the rules conflict, following the alternative rule means that we are either taking a negative NPV investment or turning down a positive NPV investment. In these cases, the alternative rules lead to bad decisions that reduce wealth.

¹ "The Theory and Practice of Corporate Finance: Evidence from the Field," *Journal of Financial Economics* 60 (2001): 187–243.

² "A Survey of Capital Budgeting Techniques Used by Major U.S. Firms," *Financial Management* 6 (1977): 66–71.

CONCEPT CHECK

1. Explain the NPV rule for stand-alone projects.
2. What does the difference between the cost of capital and the IRR indicate?

INTERVIEW WITH
DICK GRANNIS

Dick Grannis was Senior Vice President and Treasurer of QUALCOMM Incorporated, a world leader in digital wireless communications technology and semiconductors, headquartered in San Diego. He oversaw the company's \$10 billion cash investment portfolio and worked on major decisions involving acquisitions, capital structure, and international finance.

QUESTION: *QUALCOMM has a wide variety of products in different business lines. How does your capital budgeting process for new products work?*

ANSWER: QUALCOMM evaluates new projects (such as new products, equipment, technologies, research and development, acquisitions, and strategic investments) by using traditional financial measurements including discounted cash flow/NPV models, IRR levels, peak funding requirements, the time needed to reach cumulative positive cash flows, and the short-term impact of the investment on our reported net earnings. For strategic investments, we consider the possible value of financial, competitive, technology and/or market value enhancements to our core businesses—even if those benefits cannot be quantified. Overall, we make capital budgeting decisions based on a combination of objective analyses and our own business judgment.

We do not engage in capital budgeting and analysis if the project represents an immediate and necessary requirement for our business operations. One example is new software or production equipment to start a project that has already received approval.

We are also mindful of the opportunity costs of allocating our internal engineering resources on one project vs. another project. We view this as a constantly challenging but worthwhile exercise, because we have many attractive opportunities but limited resources to pursue them.

QUESTION: *How often does QUALCOMM evaluate its hurdle rates and what factors does it consider in setting them? How do you allocate capital across areas and regions and assess the risk of non-U.S. investments?*

ANSWER: QUALCOMM encourages its financial planners to utilize hurdle (or discount) rates that vary according to the risk of the particular project. We expect a rate of return commensurate with the project's risk. Our finance staff considers a wide range of discount rates and chooses one that fits the project's expected risk profile and time horizon. The range can be from 6% to 8% for relatively safe investments in the domestic market to 50% or more for equity investments in foreign markets that may be illiquid and difficult to predict. We re-evaluate our hurdle rates at least every year.

We analyze key factors including: (1) market adoption risk (whether or not customers will buy the new product or service at the price and volume we expect), (2) technology development risk (whether or not we can develop and patent the new product or service as expected), (3) execution risk (whether we can launch the new product or service cost effectively and on time), and (4) dedicated asset risk (the amount of resources that must be consumed to complete the work).

QUESTION: *How are projects categorized and how are the hurdle rates for new projects determined? What would happen if QUALCOMM simply evaluated all new projects against the same hurdle rate?*

ANSWER: We primarily categorize projects by risk level, but we also categorize projects by the expected time horizon. We consider short-term and long-term projects to balance our needs and achieve our objectives. For example, immediate projects and opportunities may demand a great amount of attention, but we also stay focused on long-term projects because they often create greater long-term value for stockholders.

If we were to evaluate all new projects against the same hurdle rate, then our business planners would, by default, consistently choose to invest in the highest risk projects because those projects would appear to have the greatest expected returns in DCF models or IRR analyses. That approach would probably not work well for very long

7.2 The Internal Rate of Return Rule

One interpretation of the internal rate of return is the average return earned by taking on the investment opportunity. The internal rate of return (IRR) investment rule is based on this idea. If the average return on the investment opportunity (i.e., the IRR) is greater than the return on other alternatives in the market with equivalent risk and maturity (i.e., the project's cost of capital), you should undertake the investment opportunity. We state the rule formally as follows:

IRR Investment Rule: Take any investment opportunity where the IRR exceeds the opportunity cost of capital. Turn down any opportunity whose IRR is less than the opportunity cost of capital.

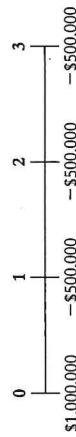
Applying the IRR Rule

Like the NPV rule, the internal rate of return investment rule is applied to single, stand-alone projects within the firm. The IRR investment rule will give the correct answer (that is, the same answer as the NPV rule) in many—but not all—situations. For instance, it gives the correct answer for Fredrick's fertilizer opportunity. Looking again at Figure 7.1, whenever the cost of capital is below the IRR (14%), the project has a positive NPV and you should undertake the investment.

In the Fredrick fertilizer example, the NPV rule and the IRR rule coincide, so the IRR rule gives the correct answer. This need not always be the case, however. In fact, the IRR rule is only guaranteed to work for a stand-alone project if all of the project's negative cash flows precede its positive cash flows. If this is not the case, the IRR rule can lead to incorrect decisions. Let's examine several situations in which the IRR fails.

Pitfall #1: Delayed Investments

John Star, the founder of SuperTech, the most successful company in the last 20 years, has just retired as CEO. A major publisher has offered to pay Star \$1 million up front if he agrees to write a book about his experiences. He estimates that it will take him three years to write the book. The time that he spends writing will cause him to forgo alternative sources of income amounting to \$500,000 per year. Considering the risk of his alternative income sources and available investment opportunities, Star estimates his opportunity cost of capital to be 10%. The timeline of Star's investment opportunity is



The NPV of Star's investment opportunity is

$$NPV = 1,000,000 - \frac{500,000}{1+r} - \frac{500,000}{(1+r)^2} - \frac{500,000}{(1+r)^3}$$

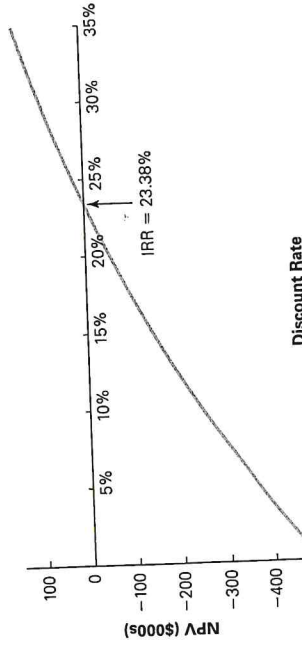
By setting the NPV equal to zero and solving for r , we find the IRR. Using the annuity spreadsheet:

	NPV	RATE	PV	PMT	FV	Excel Formula
Given	3		1,000,000	-500,000	0	
Solve for	1	23.38%				=RATE(3, -500000, 1000000, 0)

FIGURE 7.2

NPV of Star's \$1 Million Book Deal

When the benefits of an investment occur before the costs, the NPV is an increasing function of the discount rate, and the IRR rule fails.



The 23.38% IRR is larger than the 10% opportunity cost of capital. According to the IRR rule, Star should sign the deal. But what does the NPV rule say?

$$NPV = 1,000,000 - \frac{500,000}{1.1} - \frac{500,000}{1.1^2} - \frac{500,000}{1.1^3} = -\$243,426$$

At a 10% discount rate, the NPV is negative, so signing the deal would reduce Star's wealth. He should not sign the book deal.

To understand why the IRR rule fails, Figure 7.2 shows the NPV profile of the book deal. No matter what the cost of capital is, the IRR rule and the NPV rule will give exactly opposite recommendations. That is, the NPV is positive only when the opportunity cost of capital is above 23.38% (the IRR). In fact, Star should accept the investment only when the opportunity cost of capital is greater than the IRR, the opposite of what the IRR rule recommends.

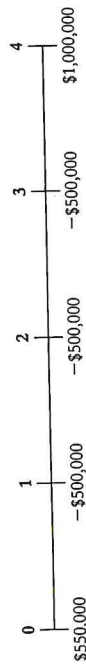
Figure 7.2 also illustrates the problem with using the IRR rule in this case. For most investment opportunities, expenses occur initially and cash is received later. In this case, Star gets cash up front and incurs the costs of producing the book later. It is as if Star borrowed money—receiving cash today in exchange for a future liability—and when you borrow money you prefer as low a rate as possible. In this case the IRR is best interpreted as the rate Star is paying rather than earning, and so Star's optimal rule is to borrow money so long as this rate is less than his cost of capital.

Even though the IRR rule fails to give the correct answer in this case, the IRR itself still provides useful information in conjunction with the NPV rule. As mentioned earlier, IRR indicates how sensitive the investment decision is to uncertainty in the cost of capital estimate. In this case, the difference between the cost of capital and the IRR is large—13.38%. Star would have to have underestimated the cost of capital by 13.38% to make the NPV positive.

Pitfall #2: Multiple IRRs

Star has informed the publisher that it needs to sweeten the deal before he will accept it. In response, the publisher offers to give him a royalty payment when the book is published in exchange for taking a smaller up-front payment. Specifically, Star will receive \$1 million when the book is published and sold four years from now, together with an up-front payment of \$550,000. Should he accept or reject the new offer?

We begin with the new timeline:



The NPV of Star's new offer is

$$NPV = 550,000 - \frac{500,000}{1+r} - \frac{500,000}{(1+r)^2} - \frac{500,000}{(1+r)^3} + \frac{1,000,000}{(1+r)^4}$$

By setting the NPV equal to zero and solving for r , we find the IRR. In this case, there are *two* IRRs—that is, there are two values of r that set the NPV equal to zero. You can verify this fact by substituting IRRs of 7.164% and 33.673% into the equation. Because there is more than one IRR, we cannot apply the IRR rule.

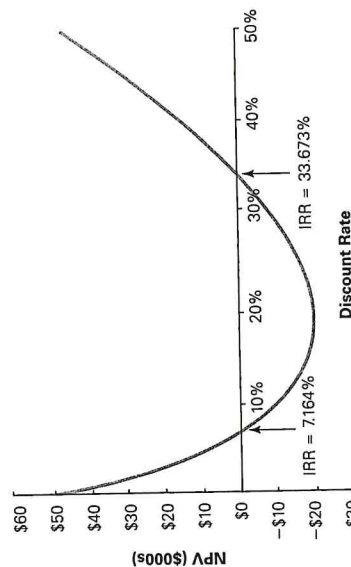
For guidance, let's turn to the NPV rule. Figure 7.3 shows the NPV profile of the new offer. If the cost of capital is *either* below 7.164% or above 33.673%, Star should undertake the opportunity. Otherwise, he should turn it down. Notice that even though the IRR rule fails in this case, the two IRRs are still useful as bounds on the cost of capital. If the cost of capital estimate is wrong, and it is actually smaller than 7.164% or larger than 33.673%, the decision not to pursue the project will change. Even if Star is uncertain whether his actual cost of capital is 10%, as long as he believes it is within these bounds, he can have a high degree of confidence in his decision to reject the deal.

There is no easy fix for the IRR rule when there are multiple IRRs. Although the NPV is negative between the IRRs in this example, the reverse is also possible. Furthermore, there are situations in which more than two IRRs exist.³ When multiple IRRs exist, our only choice is to rely on the NPV rule.

FIGURE 7.3

NPV of Star's Book Deal with Royalties

In this case, there is more than one IRR, invalidating the IRR rule. In this case, Star should only take the offer if the opportunity cost of capital is *either* below 7.164% or above 33.673%.



³ In general, there can be as many IRRs as the number of times the project's cash flows change sign over time.

COMMON MISTAKE

The examples in this section illustrate the potential shortcomings of the IRR rule when choosing to accept or reject a stand-alone project. As we said at the outset, we can only avoid these problems if all of the negative cash flows of the project precede the positive cash flows. Otherwise, we cannot rely on the IRR rule. However, even in that case,

the IRR itself remains a very useful tool. The IRR measures the average return over the life of an investment and indicates the sensitivity of the NPV to estimation error in the cost of capital. Thus, knowing the IRR can be very useful, but relying on it alone to make investment decisions can be hazardous.

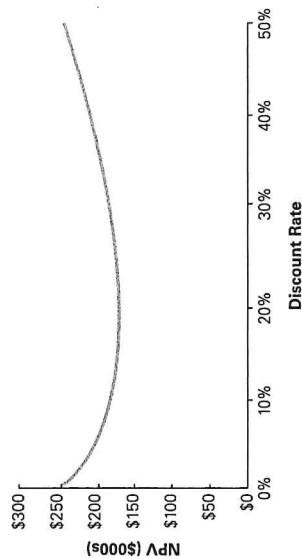
Pitfall #3: Nonexistent IRR

After protracted negotiations, Star is able to get the publisher to increase his initial payment to \$750,000, in addition to his \$1 million royalty payment when the book is published in four years. With these cash flows, no IRR exists; that is, there is no discount rate that makes the NPV equal to zero. Thus, the IRR rule provides no guidance whatsoever. To evaluate this final offer, let's again look at the NPV profile, shown in Figure 7.4. There we can see that the NPV is positive for any discount rate, and so the offer is attractive. But don't be fooled into thinking the NPV is always positive when the IRR does not exist—it can just as well be negative.

FIGURE 7.4

NPV of Star's Final Offer

In this case, the NPV is positive for every discount rate, and so there is no IRR. Thus, we cannot use the IRR rule.



EXAMPLE 7.1

Problems with the IRR Rule

Problem

Consider projects with the following cash flows:

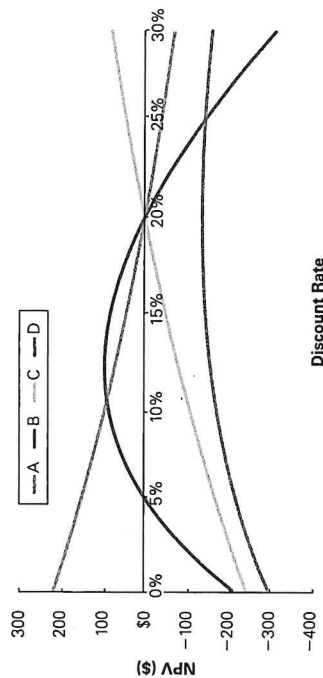
Project	0	1	2
A	-375	-300	900
B	-22,222	50,000	-28,000
C	400	400	-1,056
D	-4,300	10,000	-6,000

Which of these projects have an IRR close to 20%? For which of these projects does the IRR rule provide the correct decision?

Solution

We plot the NPV profile for each project below. From the NPV profiles, we can see that projects A, B, and C each have an IRR of approximately 20%, while project D has no IRR. Note also that project B has another IRR of 5%.

The IRR rule provides the correct decision only if the project has a positive NPV for every discount rate below the IRR. Thus, the IRR rule is only valid for project A. This project is the only one for which all the negative cash flows precede the positive ones.



An interactive tool that allows you to compare NPV profiles for alternative cash flows is available in the eTextbook and MyLab Finance.

As the previous examples demonstrate, whenever a project has positive cash flows that precede negative ones, looking at the project's NPV profile is necessary to interpret the IRR. The appendix to this chapter explains how to calculate the NPV profile in Excel.

CONCEPT CHECK

1. Under what conditions do the IRR rule and the NPV rule coincide for a stand-alone project?
2. If the IRR rule and the NPV rule lead to different decisions for a stand-alone project, which should you follow? Why?

7.3 The Payback Rule

In this section, we examine the *payback rule* as an alternative decision rule for single, stand-alone projects within the firm. The **payback investment rule** states that you should only accept a project if its cash flows pay back its initial investment within a prespecified period. To apply the payback rule, you first calculate the amount of time it takes to pay back the initial investment, called the **payback period**. Then you accept the project if the payback period is less than a prespecified length of time—usually a few years. Otherwise, you reject the project. For example, a firm might adopt any project with a payback period of less than two years.

Applying the Payback Rule

To illustrate the payback rule, we return to the Fredrick's Feed and Farm example.

EXAMPLE 7.2 The Payback Rule

Problem

Assume Fredrick's requires all projects to have a payback period of five years or less. Would the firm undertake the fertilizer project under this rule?

Solution

Recall that the project requires an initial investment of \$250 million, and will generate \$35 million per year. The sum of the cash flows from year 1 to year 5 is $\$35 \times 5 = \175 million, which will not cover the initial investment of \$250 million. In fact, it will not be until year 8 that the initial investment will be paid back ($\$35 \times 8 = \280 million). Because the payback period for this project exceeds five years, Fredrick's will reject the project.

Relying on the payback rule analysis in Example 7.2, Fredrick's will reject the project. However, as we saw earlier, with a cost of capital of 10%, the NPV is \$100 million. Following the payback rule would be a mistake because Fredrick's would pass up a project worth \$100 million.

Payback Rule Pitfalls in Practice

The payback rule is not as reliable as the NPV rule because it (1) ignores the project's cost of capital and the time value of money, (2) ignores cash flows after the payback period, and (3) relies on an ad hoc decision criterion (what is the right number of years to require for the payback period?).⁴ Despite these failings, about 57% of the firms Graham and Harvey surveyed reported using the payback rule as part of the decision-making process.

Why do some companies consider the payback rule? The answer probably relates to its simplicity. This rule is typically used for small investment decisions—for example, whether to purchase a new copy machine or to service the old one. In such cases, the cost of making an incorrect decision might not be large enough to justify the time required to calculate the NPV. The payback rule also provides budgeting information regarding the length of time capital will be committed to a project. Some firms are unwilling to commit capital to long-term investments without greater scrutiny. Also, if the required payback period is short (one or two years), then most projects that satisfy the payback rule will have a positive NPV. So firms might save effort by first applying the payback rule, and only if it fails take the time to compute NPV.

CONCEPT CHECK

1. Can the payback rule reject projects that have positive NPV? Can it accept projects that have negative NPV?
2. If the payback rule does not give the same answer as the NPV rule, which rule should you follow? Why?

⁴ Some companies address the first failing by computing the payback period using discounted cash flows (called discounted payback).

Why Do Rules Other Than the NPV Rule Persist?

Professors Graham and Harvey found that a sizable minority of firms (25%) in their study do not use the NPV rule at all. In addition, more than half of firms surveyed used the payback rule. Furthermore, it appears that most firms use *both* the NPV rule and the IRR rule. Why do firms use rules other than NPV if they can lead to erroneous decisions?

One possible explanation for this phenomenon is that Graham and Harvey's survey results might be misleading. Managers may use the payback rule for budgeting purposes or as a shortcut to get a quick sense of the project before calculating NPV. Similarly, CFOs who were using the IRR as a sensitivity measure in conjunction with the NPV rule might have checked both the IRR box and the NPV box on the survey. Nevertheless, a significant minority of managers surveyed replied that they used only the IRR rule, so this explanation cannot be the whole story.

Managers may use the IRR rule exclusively because you do not need to know the opportunity cost of capital to calculate the IRR. But this benefit is superficial: While you may not need to know the cost of capital to *calculate* the IRR, you certainly need to know the cost of capital when you *apply* the IRR rule. Consequently, the opportunity cost of capital is as important to the IRR rule as it is to the NPV rule.

Nonetheless, part of the appeal of the IRR rule is that the IRR appears to sum up the attractiveness of an investment without requiring an assumption about the cost of capital. However, a more useful summary is the project's NPV profile, showing the NPV as a function of the discount rate. The NPV profile also does not require knowing the cost of capital, and it has the distinct advantage of being much more informative and reliable.

7.4 Choosing between Projects

Thus far, we have considered only decisions where the choice is either to accept or to reject a single, stand-alone project. Sometimes, however, a firm must choose just one project from among several possible projects, that is, the choices are mutually exclusive. For example, a manager may be evaluating alternative package designs for a new product. When choosing any one project excludes us from taking the others, we are facing mutually exclusive investments.

NPV Rule and Mutually Exclusive Investments

When projects are mutually exclusive, we need to determine which projects have a positive NPV and then rank the projects to identify the best one. In this situation, the NPV rule provides a straightforward answer: *Pick the project with the highest NPV*. Because the NPV expresses the value of the project in terms of cash today, picking the project with the highest NPV leads to the greatest increase in wealth.

EXAMPLE 7.3

NPV and Mutually Exclusive Projects

Problem

A small commercial property is for sale near your university. Given its location, you believe a student-oriented business would be very successful there. You have researched several possibilities and come up with the following cash flow estimates (including the cost of purchasing the property). Which investment should you choose?

Project	Initial Investment	First-Year Cash Flow	Growth Rate	Cost of Capital
Book Store	\$300,000	\$63,000	3.0%	8%
Coffee Shop	\$400,000	\$80,000	3.0%	8%
Music Store	\$400,000	\$104,000	0.0%	8%
Electronics Store	\$400,000	\$100,000	3.0%	11%

Solution

Assuming each business lasts indefinitely, we can compute the present value of the cash flows from each as a constant growth perpetuity. The NPV of each project is

$$NPV(\text{Book Store}) = -300,000 + \frac{63,000}{8\% - 3\%} = \$960,000$$

$$NPV(\text{Coffee Shop}) = -400,000 + \frac{80,000}{8\% - 3\%} = \$1,200,000$$

$$NPV(\text{Music Store}) = -400,000 + \frac{104,000}{8\%} = \$900,000$$

$$NPV(\text{Electronics Store}) = -400,000 + \frac{100,000}{11\% - 3\%} = \$850,000$$

Thus, all of the alternatives have a positive NPV. But, because we can only choose one, the coffee shop is the best alternative.

IRR Rule and Mutually Exclusive Investments

Because the IRR is a measure of the expected return of investing in the project, you might be tempted to extend the IRR investment rule to the case of mutually exclusive projects by picking the project with the highest IRR. Unfortunately, picking one project over another simply because it has a larger IRR can lead to mistakes. In particular, *when projects differ in their scale of investment, the timing of their cash flows, or their riskiness, then their IRRs cannot be meaningfully compared*.

Differences in Scale. Would you prefer a 500% return on \$1, or a 20% return on \$1 million? While a 500% return certainly sounds impressive, at the end of the day you will only make \$5. The latter return sounds much more mundane, but you will make \$200,000. This comparison illustrates an important shortcoming of IRR: Because it is a return, you cannot tell how much value will actually be created without knowing the scale of the investment.

If a project has a positive NPV, then if we can double its size, its NPV will double: By the Law of One Price, doubling the cash flows of an investment opportunity must make it worth twice as much. However, the IRR rule does not have this property—it is unaffected by the scale of the investment opportunity because the IRR measures the average return of the investment. Hence, we cannot use the IRR rule to compare projects of different scales.

As an illustration of this situation, consider the investment in the book store versus the coffee shop in Example 7.3. We can compute the IRR of each as follows:

$$\text{Book Store: } -300,000 + \frac{63,000}{IRR - 3\%} = 0 \Rightarrow IRR = 24\%$$

$$\text{Coffee Shop: } -400,000 + \frac{80,000}{IRR - 3\%} = 0 \Rightarrow IRR = 23\%$$

Both projects have IRRs that exceed their cost of capital of 8%. But although the coffee shop has a lower IRR, because it is on a larger scale of investment (\$400,000 versus \$300,000), it generates a higher NPV (\$1.2 million versus \$960,000) and thus is more valuable.

Differences in Timing. Even when projects have the same scale, the IRR may lead you to rank them incorrectly due to differences in the timing of the cash flows. The IRR is expressed as a return, but the dollar value of earning a given return—and therefore its NPV—depends on how long the return is earned. Earning a very high annual return is much more valuable if you earn it for several years than if you earn it for only a few days.

As an example, consider the following short-term and long-term projects:

Year	0	1	2	3	4	5
Short-Term Project	-100	150				
Long-Term Project	-100					

$100 \times 1.50^5 = 759.375$

Both projects have an IRR of 50%, but one lasts for one year, while the other has a five-year horizon. If the cost of capital for both projects is 10%, the short-term project has an NPV of $-100 + 150/1.10 = \$36.36$, whereas the long-term project has an NPV of $-100 + 759.375/1.10^5 = \371.51 . Notice that despite having the same IRR, the long-term project is more than 10 times as valuable as the short-term project.

Even when projects have the same horizon, the pattern of cash flows over time will often differ. Consider again the coffee shop and music store investment alternatives in Example 7.3. Both of these investments have the same initial scale, and the same horizon (infinite). The IRR of the music store investment is

$$\text{Music Store: } -400,000 + \frac{104,000}{IRR} = 0 \Rightarrow IRR = 26\%$$

But although the music store has a higher IRR than the coffee shop (26% versus 23%), it has a lower NPV (\$900,000 versus \$1.2 million). The reason the coffee shop has a higher NPV despite having a lower IRR is its higher growth rate. The coffee shop has lower initial cash flows but higher long-run cash flows than the music store. The fact that its cash flows are relatively delayed makes the coffee shop effectively a longer-term investment.

Differences in Risk. To know whether the IRR of a project is attractive, we must compare it to the project's cost of capital, which is determined by the project's risk. Thus, an IRR that is attractive for a safe project need not be attractive for a risky project. As a simple example, while you might be quite pleased to earn a 10% return on a risk-free investment opportunity, you might be much less satisfied to earn a 10% expected return on an investment in a risky start-up company. Ranking projects by their IRRs ignores risk differences.

Looking again at Example 7.3, consider the investment in the electronics store. The IRR of the electronics store is

$$\text{Electronics Store: } -400,000 + \frac{100,000}{IRR - 3\%} = 0 \Rightarrow IRR = 28\%$$

This IRR is higher than those of all the other investment opportunities. Yet the electronics store has the lowest NPV. In this case, the investment in the electronics store is riskier, as evidenced by its higher cost of capital. Despite having a higher IRR, it is not sufficiently profitable to be as attractive as the safer alternatives.

Key Takeaways. As the above examples highlight, one should always avoid comparing IRRs when making real investment decisions. In addition to the above problems, IRRs can be easily manipulated by changing the project's financing — see the Common Mistake box on page 261. Nevertheless, errors based on a comparison of IRRs are routinely made in practice. Always be sure to compare NPVs to determine which option truly creates the most value!

The Incremental IRR

When choosing between two projects, an alternative to comparing their IRRs is to compute the **incremental IRR**, which is the IRR of the incremental cash flows that would result from replacing one project with the other. The incremental IRR tells us the discount rate

COMMON MISTAKE

Manipulating the IRR with Financing

Because the IRR is not itself a measure of value, it is easy to manipulate by restructuring the project's cash flows. In particular, it is easy to increase the IRR of a project by financing a portion of the initial investment. A common mistake in practice is to regard this higher IRR as an indication that the financing is attractive. For example, consider an investment in new equipment that will have the following cash flows:

Year	0	1
Cash Flow	-100	130

This investment has an IRR of 30%. Now suppose that seller of the equipment offers to lend us \$80, so that we only need to pay \$20 initially. In exchange, we must pay \$100 in one year. By financing the project in this way, the cash flows become

Year	0	1
Cash Flow	-20	30

The project's IRR is now $(30/20) - 1 = 50\%$. Does this higher IRR mean that the project is now more attractive? In other words, is the financing a good deal?

The answer is no. Remember, we cannot compare IRRs, so a 50% IRR is not necessarily better than a 30% IRR. In this case, the project with financing is a much smaller scale investment than without financing. In addition, borrowing money is likely to increase the risk to shareholders from underrating the project. We'll see explicitly the effect of leverage on risk and shareholders' required return in Parts 4 and 5 of the book (see e.g. Section 14.3).

In this particular example, note that we borrowed \$80 initially in exchange for paying \$100 in one year. The IRR of this loan is $(100/80) - 1 = 25\%$ (this is also the incremental IRR of rejecting the financing). This rate is probably much higher than our firm's borrowing cost if it borrowed through other means. If so, including this financing with the project would be a mistake, despite the higher IRR. We can avoid such mistakes by correctly applying the NPV rule when comparing alternatives.

at which it becomes profitable to switch from one project to the other. Then, rather than compare the projects directly, we can evaluate the decision to switch from one to the other using the IRR rule, as in the following example.

EXAMPLE 7.4

Using the Incremental IRR to Compare Alternatives

Problem

Your firm is considering overhauling its production plant. The engineering team has come up with two proposals, one for a minor overhaul and one for a major overhaul. The two options have the following cash flows (in millions of dollars):

Proposal	0	1	2	3
Minor Overhaul	-10	6	6	6
Major Overhaul	-50	25	25	25

What is the IRR of each proposal? What is the incremental IRR? If the cost of capital for both of these projects is 12%, what should your firm do?

Solution

We can compute the IRR of each proposal using the annuity calculator. For the minor overhaul, the IRR is 36.3%:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	3		-10	6	0	
Solve for Rate		36.3%				=RATE(3,6,-10,0)

For the major overhaul, the IRR is 23.4%:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	3		-50	25	0	
Solve for Rate		23.4%				=RATE(3,25,-50,0)

Which project is best? Because the projects have different scales, we cannot compare their IRRs directly. To compute the incremental IRR of switching from the minor overhaul to the major overhaul, we first compute the incremental cash flows:

Proposal	0	1	2	3
Major Overhaul	-50	25	25	25
Less: Minor Overhaul	-(-10)	-6	-6	-6
Incremental Cash Flow	-40	19	19	19

These cash flows have an IRR of 20.0%:

	NPER	RATE	PV	PMT	FV	Excel Formula
Given	3		-40	19	0	
Solve for Rate		20.0%				=RATE(3,19,-40,0)

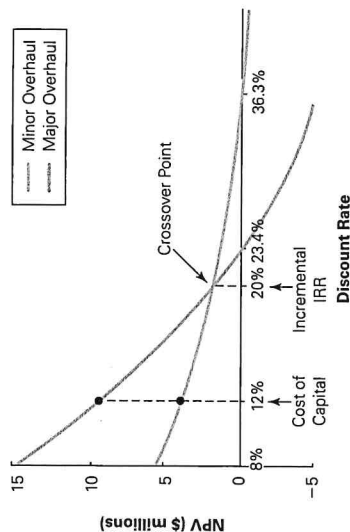
Because the incremental IRR exceeds the 12% cost of capital, switching to the major overhaul looks attractive (i.e., its larger scale is sufficient to make up for its lower IRR). We can check this result using Figure 7.5, which shows the NPV profiles for each project. At the 12% cost of capital, the NPV of the major overhaul does indeed exceed that of the minor overhaul, despite its lower IRR. Note also that the incremental IRR determines the crossover point of the NPV profiles, the discount rate for which the best project choice switches from the major overhaul to the minor one.

FIGURE 7.5

Comparison of Minor and Major Overhauls

Comparing the NPV profiles of the alternatives in Example 7.4, we can see that despite its lower IRR, the major overhaul has a higher NPV at the cost of capital of 12%. The incremental IRR of 20% is the crossover point at which the optimal decision changes.

An interactive tool that allows you to compare NPV profiles for alternative cash flows is available in the eTextbook and MyLab Finance.



As we saw in Example 7.4, the incremental IRR identifies the discount rate at which the optimal decision changes. However, when using the incremental IRR to choose between projects, we encounter all of the same problems that arose with the IRR rule:

- Even if the negative cash flows precede the positive ones for the individual projects, it need not be true for the incremental cash flows. If not, the incremental IRR is difficult to interpret, and may not exist or may not be unique.
- The incremental IRR can indicate whether it is profitable to switch from one project to another, but it does not indicate whether either project has a positive NPV on its own.

When Can Returns Be Compared?

In this chapter, we have highlighted the many pitfalls that arise when attempting to compare the IRRs of different projects. But there are many situations in which it is quite reasonable to compare returns. For example, if we were thinking of saving money in a savings account for the next year, we would likely compare the effective annual rates associated with different accounts and choose the highest option.

When is it reasonable to compare returns in this way? Remember, we can only compare returns if the investments (1) have the same scale, (2) have the same timing, and (3) have the same risk.

- When the individual projects have different costs of capital, it is not obvious what cost of capital the incremental IRR should be compared to. In this case only the NPV rule, which allows each project to be discounted at its own cost of capital, will give a reliable answer.

In summary, although the incremental IRR provides useful information by telling us the discount rate at which our optimal project choice would change, using it as a decision rule is difficult and error prone. It is much simpler to use the NPV rule.

CONCEPT CHECK

1. For mutually exclusive projects, explain why picking one project over another because it has a larger IRR can lead to mistakes.
2. What is the incremental IRR and what are its shortcomings as a decision rule?

7.5 Project Selection with Resource Constraints

In principle, the firm should take on all positive-NPV investments it can identify. In practice, there are often limitations on the number of projects the firm can undertake. For example, when projects are mutually exclusive, the firm can only take on one of the projects even if many of them are attractive. Often this limitation is due to resource constraints—for example, there is only one property available in which to open either a coffee shop, or book store, and so on. Thus far, we have assumed that the different projects the firm is considering have the same resource requirements (in Example 7.3, each project would use 100% of the property). In this section, we develop an approach for situations where the choices have differing resource needs.

Evaluating Projects with Different Resource Requirements

In some situations, different projects will demand different amounts of a particular scarce resource. For example, different products may consume different proportions of a firm's production capacity, or might demand different amounts of managerial time and attention. If there is a fixed supply of the resource so that you cannot undertake all possible opportunities, then the firm must choose the best set of investments it can make given the resources it has available.

Often, individual managers work within a budget constraint that limits the amount of capital they may invest in a given period. In this case, the manager's goal is to choose the projects that maximize the total NPV while staying within her budget. Suppose you are considering the three projects shown in Table 7.1. Absent any budget constraint, you would invest in all of these positive-NPV projects. Suppose, however, that you have a budget of at most \$100 million to invest. While Project I has the highest NPV, it uses up the entire budget. Projects II and III can both be undertaken (together they also take up the entire budget), and their combined NPV exceeds the NPV of Project I. Thus, with a budget of \$100 million, the best choice is to take Projects II and III for a combined NPV of \$130 million, compared to just \$110 million for Project I alone.

Profitability Index

Note that in the last column of Table 7.1 we included the ratio of the project's NPV to its initial investment. This ratio tells us that for every dollar invested in Project I, we will generate \$1.10 in value (over and above the dollar invested).⁵ Both Projects II and III generate higher NPVs per dollar invested than Project I, which indicates that they will use the available budget more efficiently.

In this simple example, identifying the optimal combination of projects to undertake is straightforward. In actual situations replete with many projects and resources, finding the optimal combination can be difficult. Practitioners often use the **profitability index** to identify the optimal combination of projects to undertake in such situations:

Profitability Index

$$\text{Profitability Index} = \frac{\text{Value Created}}{\text{Resource Consumed}} = \frac{\text{NPV}}{\text{Resource Consumed}} \quad (7.2)$$

The profitability index measures the “bang for your buck”—that is, the value created in terms of NPV per unit of resource consumed. After computing the profitability index, we can rank projects based on it. Starting with the project with the highest index, we move down the ranking, taking all projects until the resource is consumed. In Table 7.1, the ratio we computed in the last column is the profitability index when investment dollars are the scarce resource. Note how the “profitability index rule” would correctly select Projects II and III. We can also apply this rule when other resources are scarce, as shown in Example 7.5.

TABLE 7.1 Possible Projects for a \$100 Million Budget

Project	NPV (\$ millions)	Initial Investment (\$ millions)	Profitability Index NPV/Investment
I	110	100	1.1
II	70	50	1.4
III	60	50	1.2

⁵ Practitioners sometimes add 1 to this ratio to include the dollar invested (i.e., Project I generates a total of \$2.10 per dollar invested, generating \$1.10 in new value). Leaving out the 1 and just considering the *net* present value allows the ratio to be applied to other resources besides cash budgets, as shown in Example 7.5.

EXAMPLE 7.5

Profitability Index with a Human Resource Constraint

Problem

Your division at NetIt, a large networking company, has put together a project proposal to develop a new home networking router. The expected NPV of the project is \$17.7 million, and the project will require 50 software engineers. NetIt has a total of 190 engineers available, and the router project must compete with the following other projects for these engineers:

Project	NPV (\$ millions)	Engineering Headcount
Router	17.7	50
Project A	22.7	47
Project B	8.1	44
Project C	14.0	40
Project D	11.5	61
Project E	20.6	58
Project F	12.9	32
Total	107.5	332

How should NetIt prioritize these projects?

Solution

The goal is to maximize the total NPV we can create with 190 engineers (at most). We compute the profitability index for each project, using Engineering Headcount in the denominator, and then sort projects based on the index:

Project	NPV (\$ millions)	Engineering Headcount (EHC)	Profitability Index (NPV per EHC)	Cumulative EHC Required
Project A	22.7	47	0.483	47
Project F	12.9	32	0.403	79
Project E	20.6	58	0.355	137
Router	17.7	50	0.354	187
Project C	14.0	40	0.350	
Project D	11.5	61	0.189	
Project B	8.1	44	0.184	

We now assign the resource to the projects in descending order according to the profitability index. The final column shows the cumulative use of the resource as each project is taken on until the resource is used up. To maximize NPV within the constraint of 190 engineers, NetIt should choose the first four projects on the list. There is no other combination of projects that will create more value without using more engineers than we have. Note, however, that the resource constraint forces NetIt to forgo three otherwise valuable projects (C, D, and B) with a total NPV of \$33.6 million.

Note that in the above examples, the firm's resource constraints cause it to pass up positive-NPV projects. The highest profitability index available from these remaining projects provides useful information regarding the value of that resource to the firm. In Example 7.5, for example, Project C would generate \$350,000 in NPV per engineer. If the firm could recruit and train new engineers at a cost of less than \$350,000 per engineer, it would be worthwhile to do so in order to undertake Project C. Alternatively, if Engineering Headcount has been allocated to another division of the firm for projects with a profitability index of less than \$350,000 per engineer, it would be worthwhile to reallocate that headcount to this division to undertake Project C.

Shortcomings of the Profitability Index

Although the profitability index is simple to compute and use, for it to be completely reliable, two conditions must be satisfied:

1. The set of projects taken following the profitability index ranking completely exhausts the available resource.
2. There is only a single relevant resource constraint.

To see why the first condition is needed, suppose in Example 7.5 that NetIt has an additional small project with an NPV of only \$120,000 that requires three engineers. The profitability index in this case is $0.12/3 = 0.04$, so this project would appear at the bottom of the ranking. However, notice that three of the 190 employees are not being used after the first four projects are selected. As a result, it would make sense to take on this project even though it would be ranked last. This shortcoming can also affect highly ranked projects. For example, in Table 7.1, suppose Project III had an NPV of only \$25 million, making it significantly worse than the other projects. Then the best choice would be Project I even though Project II has a higher profitability index.

In many cases, the firm may face multiple resource constraints. For instance, there may be a budget limit as well as a headcount constraint. If more than one resource constraint is binding, then there is no simple index that can be used to rank projects. Instead, linear and integer programming techniques have been developed specifically to tackle this kind of problem. Even if the set of alternatives is large, by using these techniques on a computer we can readily calculate the set of projects that will maximize the total NPV while satisfying multiple constraints (see Further Reading for references).

CONCEPT CHECK

1. Explain why ranking projects according to their NPV might not be optimal when you evaluate projects with different resource requirements.
2. How can the profitability index be used to identify attractive projects when there are resource constraints?

Key Points and Equations

7.1 NPV and Stand-Alone Projects

- If your objective is to maximize wealth, the NPV rule always gives the correct answer.
- The difference between the cost of capital and the IRR is the maximum amount of estimation error that can exist in the cost of capital estimate without altering the original decision.

7.2 The Internal Rate of Return Rule

- IRR investment rule: Take any investment opportunity whose IRR exceeds the opportunity cost of capital. Turn down any opportunity whose IRR is less than the opportunity cost of capital.
- Unless all of the negative cash flows of the project precede the positive ones, the IRR rule may give the wrong answer and should not be used. Furthermore, there may be multiple IRRs or the IRR may not exist.

7.3 The Payback Rule

- Payback investment rule: Calculate the amount of time it takes to pay back the initial investment (the payback period). If the payback period is less than a prespecified length of time, accept the project. Otherwise, turn it down.
- The payback rule is simple, and favors short-term investments. But it is often incorrect.

7.4 Choosing between Projects

- When choosing among mutually exclusive investment opportunities, pick the opportunity with the highest NPV.
- We cannot use the IRR to compare investment opportunities unless the investments have the same scale, timing, and risk.
- Incremental IRR: When comparing two mutually exclusive opportunities, the incremental IRR is the IRR of the difference between the cash flows of the two alternatives. The incremental IRR indicates the discount rate at which the optimal project choice changes.

7.5 Project Selection with Resource Constraints

- When choosing among projects competing for the same resource, rank the projects by their profitability indices and pick the set of projects with the highest profitability indices that can still be undertaken given the limited resource.

$$\text{Profitability Index} = \frac{\text{Value Created}}{\text{Resource Consumed}} = \frac{\text{NPV}}{\text{Resource Consumed}} \quad (7.2)$$

- The profitability index is only completely reliable if the set of projects taken following the profitability index ranking completely exhausts the available resource and there is only a single relevant resource constraint.

Key Terms

Data Table p. 274	NPV profile p. 249
incremental IRR p. 260	payback investment rule p. 256
internal rate of return (IRR)	payback period p. 256
investment rule p. 252	profitability index p. 264

Further Reading

Readers who would like to know more about what managers actually do should consult J. Graham and C. Harvey, "How CFOs Make Capital Budgeting and Capital Structure Decisions," *Journal of Applied Corporate Finance* 15 (2002): 8–23; S. H. Kim, T. Crick, and S. H. Kim, "Do Executives Practice What Academics Preach?" *Management Accounting* 68 (1986): 49–52; and P. Ryan and G. Ryan, "Capital Budgeting Practices of the Fortune 1000: How Have Things Changed?" *Journal of Business and Management* 8 (2002): 355–364.

Numerous approaches have been developed to fix some of the shortcomings of the internal rate of return, such as the modified IRR (MIRR) and the average IRR (AIRR). For a survey of the different approaches, and analysis showing how AIRR can be made consistent with NPV, see C. Magni, "Capital Depreciation and the Underdetermination of Rate of Return: A Unifying Perspective," *Journal of Mathematical Economics* 67 (2010): 54–79.

For readers interested in how to select among projects competing for the same set of resources, the following references will be helpful: M. Vanhoucke, E. Demeulemeester, and W. Herroelen, "On Maximizing the Net Present Value of a Project Under Renewable Resource Constraints," *Management Science* 47 (2001): 1113–1121; and H. M. Weingartner, *Mathematical Programming and the Analysis of Capital Budgeting Problems* (Englewood Cliffs, NJ: Prentice-Hall, 1963).

Problems NPV and Stand-Alone Projects

1. Your brother wants to borrow \$10,000 from you. He has offered to pay you back \$12,000 in a year. If the cost of capital of this investment opportunity is 10%, what is its NPV? Should you undertake the investment opportunity? Calculate the IRR and use it to determine the maximum deviation allowable in the cost of capital estimate to leave the decision unchanged.
2. You are considering investing in a start-up company. The founder asked you for \$200,000 today and you expect to get \$1,000,000 in nine years. Given the riskiness of the investment opportunity, your cost of capital is 20%. What is the NPV of the investment opportunity? Should you undertake the investment opportunity? Calculate the IRR and use it to determine the maximum deviation allowable in the cost of capital estimate to leave the decision unchanged.